

Thai Viet Nguyen

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SUMMARY

Full-Stack Engineer with 3+ years of experience building scalable Node.js and TypeScript applications, REST APIs, and microservices for enterprise and startup platforms. Experienced in AI-enabled services integrating Azure OpenAI, Python microservices, messaging systems, and real-time communication, with strengths in API security, testing, observability, and cloud deployment. Skilled in designing reliable backend systems across the full delivery lifecycle.

TECHNICAL SKILLS

- **Core:** JavaScript, TypeScript, Node.js, SQL, NoSQL, Python
- **Frameworks & Architecture:** Express.js, NestJS, React.js, FastAPI, Microservices, REST APIs, GraphQL, Message Queues, Socket.IO
- **Data & Messaging:** PostgreSQL, MySQL, MongoDB, Redis, Kafka, RabbitMQ, Docker
- **Cloud:** AWS, Azure (working knowledge of core compute, storage, and deployment concepts)
- **Security:** JWT, OAuth 2.0, Role-Based Access Control (RBAC), PII-aware data handling
- **Testing:** Jest (unit testing)
- **API Design:** OpenAPI/Swagger documentation, REST versioning, request validation
- **Observability:** Datadog, Grafana, AWS CloudWatch (logging, monitoring, alerting)
- **Methodologies & Tools:** Git/GitHub, Jira, Agile/Scrum, CI/CD, SDLC
- **AI & LLM:** LLM APIs, Tool Calling, Vector Databases, Prompt Orchestration, Retrieval-Augmented Generation (RAG), Embeddings

WORK EXPERIENCE

AIVN

Nov 2024 - Aug 2026

Full-Stack Developer

Ha Noi, Vietnam

- **Early core member of the engineering team;** helped build the system architecture from the ground up and contributed to product research and direction.
- Direct experience with **AI and agentic coding tools** such as **Claude Code, OpenAI Codex, Cursor.**
- Designed and built **Node.js** and **ExpressJS REST APIs** and **ReactJS** features in TypeScript, integrating Python-based AI microservices and **LLM APIs** with **Azure OpenAI** to evaluate users for a platform serving **more than 200,000 users.**
- Implement and maintain **RabbitMQ** messaging and **Socket.IO** real-time communication within a microservices architecture, processing messages daily.
- **Write unit tests** with Jest to validate business logic and reduce regressions across core features.
- Own UI/UX, business logic, and database schema design **end-to-end for new features.**
- Lead code reviews for a team of 6 developers and manage GitHub-based CI/CD workflows **to maintain release quality and stability.**
- Troubleshoot and resolve production hotfixes and critical issues, working to minimize incident response time and modernize legacy codebases.
- Optimize core business logic and database query performance to reduce workflow processing time and improve system responsiveness.
- Integrated Retrieval-Augmented Generation (**RAG**), embeddings, vector databases, prompt orchestration, and tool calling into AI-enabled backend workflows.

VMO Holdings

Sep 2023 - Nov 2024

Back-End Developer

Hanoi, Vietnam

- Engineered **Node.js** and **ExpressJS REST APIs** for user verification and admin management within an enterprise **microservices architecture** for a US-based KYC platform.
- Implemented authentication and authorization using JWT, OAuth 2.0, and role-based access control (**RBAC**) to secure identity-verification workflows handling sensitive PII.
- Documented and maintained APIs with **OpenAPI/Swagger** to standardize request/response contracts across services.
- Monitored application health and diagnosed issues using **Datadog, Grafana, and AWS CloudWatch** for logging and alerting.
- Integrated third-party identity-verification services via REST APIs, reducing manual document review time through automated workflows.
- Optimized admin management efficiency and **SQL** data retrieval by refactoring legacy logic and improving query structures.
- Designed **NestJS APIs** for order and store management using a Kafka-based microservices architecture in an Agile environment.
- Reduced data mismatches by 15% by redesigning core order-history and stock-tracking logic for high-throughput e-commerce workflows.

FPT Software

Jun 2022 - Sep 2022

Software Development Intern

Hanoi, Vietnam

- Built front-end and back-end components using JavaScript, jQuery, and MySQL, following Software Development Lifecycle (SDLC)

standards.

- Practiced enterprise-level development methodologies and participated in technical training programs to support streamlined software delivery.

Projects

AI Sales Agent — Multi-tenant SaaS platform (*solo project, 2026*)

2026

Full-Stack Developer

- A B2B SaaS where a business configures an AI sales assistant over its own knowledge base — documents, product catalogue, versioned prompts — and that assistant answers its customers' questions with grounded, cited answers.
- Built a **provider-agnostic LLM layer** (OpenAI, Anthropic, and a network-free fake for tests) behind one interface, with per-message token and cost accounting, and token streaming to the browser over SSE.
- Implemented an end-to-end RAG pipeline: document upload → background arq worker (PDF/DOCX/HTML extraction, chunking, embedding into pgvector) → hybrid retrieval fusing vector similarity and Postgres full-text search via Reciprocal Rank Fusion → answers grounded in the tenant's own documents with citations.
- Built a multi-step agent tool-calling loop where the model decides when to search knowledge or capture a lead, with a per-turn tool registry driven by per-agent grants events streamed on the same SSE channel.
- Shipped a CI/CD pipeline on GitHub Actions deploying API to Render, web to Vercel, and migrations to Neon as a separate one-shot job, with readiness-gated rollout

EDUCATION

Harrisburg University

2026 - Present

Master of Science, Computer Science

Harrisburg, US

FPT University

Sep 2019 - Sep 2023

Bachelor of Science, Software Engineering

Hanoi, Vietnam

- **GPA:** 3.2/4

CERTIFICATIONS

- Back End Development and APIs
- Software Development Lifecycle
- Web Design for Everybody: Basics of Web Development & Coding

LANGUAGES

- English(IELTS 6.5)
- Vietnamese(Native)

AWARDS

- Top 5, "Crafting ESG Solutions on Hedera Guardian" Hackathon, 2023